

DOMAIN MAPPER

Student: Yonatan Menashe

Network Security Project - ZX305



Introduction

Domain Mapper is a Bash-based network reconnaissance tool designed to automate information gathering, service enumeration, and vulnerability assessment within an authorized network environment. The script combines multiple stages into a single workflow, allowing the user to perform network scanning, identify services, and generate organized reports.

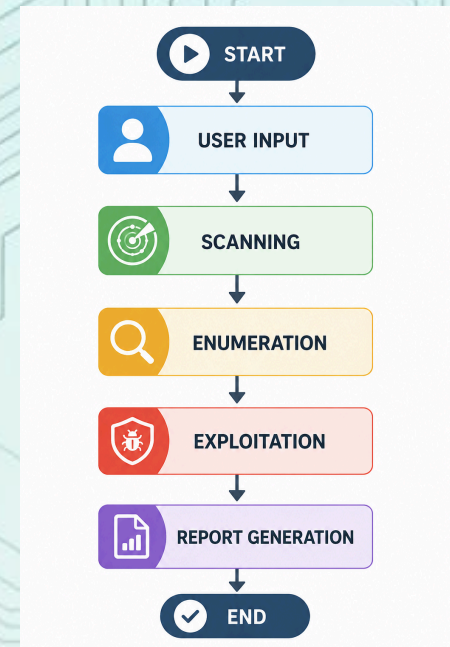
The tool supports multiple operation levels for scanning, enumeration, and exploitation, providing flexibility while maintaining a structured and easy-to-use interface. All collected information is saved in organized session folders, making the results easy to review and analyze.

Program Overview

The Domain Mapper tool is organized into several sequential stages that guide the user through the network assessment process. Each stage performs a specific task, from collecting user input to generating structured reports. This modular design improves readability, simplifies maintenance, and allows each component to operate independently.

The main stages of the program are:

- User Input Collection
- Scanning
- Enumeration
- Exploitation
- Report Generation



The selected operation level determines which functions are executed during each phase, allowing the user to perform either a quick assessment or a more comprehensive analysis.

User Input

Before starting the assessment, the program collects all required information from the user. This configuration stage ensures that the selected operations are performed according to the user's preferences and the target environment.

The collected parameters include:

- Target network range
- Domain name (optional)
- Active Directory credentials
- Password list location
- Wizard Mode selection
- Scanning level
- Enumeration level
- Exploitation level



```
(root@kali)-[/home/kali]
# ./domain_mapper.sh
[OK] Environment checks passed.

Domain Mapper
Authorized lab use only

[STEP] Input collection
Target network range (example: 192.168.56.0/24 or 127.0.0.1): 192.168.72.0/24
Domain name (example: lab.local; leave blank if unknown): netsec.local
AD username (leave blank to skip): administrator
AD password (input hidden; leave blank to skip):
Password list path [/usr/share/wordlists/rockyou.txt]: wordlists/passwords.list
Enable Wizard Mode? [y/N]: y
```

Authorization & Pre-run Summary

Before executing any network operations, the program displays an authorization warning and a summary of the selected configuration. This step allows the user to verify all settings before continuing and helps prevent accidental execution against unauthorized targets.

The summary includes the target network, domain information, provided credentials, password list, Wizard Mode status, and the selected operation levels. Only after the user confirms the configuration does the assessment begin.

[STEP] Authorization

Authorization warning:

```
Run this script only against networks you own or have explicit permission to test.
This version does not include destructive behavior, stealth, persistence,
malware, credential disclosure, password spraying, Kerberoasting, or cracking.
Exploitation Basic requires a separate exact confirmation before vuln scripts run.
```

Pre-run summary

```
Target:          192.168.72.0/24
Input domain:    netsec.local
Domain provided: yes
AD username given: yes
AD password given: yes
Password list:   wordlists/passwords.list
Wizard Mode:     yes
Scanning level:  Basic
Enumeration level: Advanced
Exploitation level: Advanced
```

```
Do you confirm this target is authorized and the selections are correct? [y/N]: y
```

Scanning

The scanning phase is responsible for discovering active hosts, identifying open ports, and detecting running services on the target network. Based on the selected operation level, the program automatically executes the appropriate Nmap scan while creating a dedicated session directory and verifying the availability of the required tools.

The generated scan results are stored for later use during the Enumeration phase, allowing the program to reuse collected information instead of performing unnecessary duplicate scans.

```
[STEP] Session setup
[OK] Created session directory: ./sessions/netsec.local_2026-07-29_13-20

[STEP] Dependency checks
[OK] Tool availability recorded.
[INFO] Starting Domain Mapper.
[INFO] Target range: 192.168.72.0/24
[INFO] Domain name: netsec.local
[INFO] AD username provided: yes
[INFO] AD password provided: yes
[INFO] Password list path: wordlists/passwords.list

[STEP] Scanning
[INFO] Starting: Basic TCP service scan (top 1000 ports)
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-29 13:20 -0400
█
```

Enumeration

The Enumeration phase analyzes the information collected during the scanning phase to identify network services and infrastructure components. Instead of repeating the service detection scan, the program reuses the previously collected scan results, reducing unnecessary network traffic and improving execution efficiency.

For the Basic level, the program performs an additional targeted UDP port 67 check to help identify potential DHCP servers, while preserving the previously collected service information.

```
[STEP] Enumeration
[INFO] Reusing service/version data from scanning output.
[INFO] Starting: Enumeration Basic DHCP UDP check
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-29 13:20 -0400
Nmap scan report for 192.168.72.1
Host is up (0.0011s latency).

PORT      STATE      SERVICE
67/udp    open|filtered dhcp
MAC Address: 00:50:56:C0:00:08 (VMware)
```

Advanced Enumeration (NSE)

The Advanced Enumeration level extends the basic enumeration process by executing targeted NSE scripts against the discovered services. These scripts collect additional information about the target system, shared resources, and the Active Directory environment.

smb-os-discovery - Collects operating system, host, and domain information.

```
[OK] Completed: Enumeration Basic DHCP UDP check  
[INFO] Refreshing parsed scan summaries.  
[INFO] Starting: NSE smb-os-discovery on discovered service hosts  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-29 13:33 -0400  
Nmap scan report for 192.168.72.154  
Host is up (0.00079s latency).
```

smb-enum-shares - Enumerates available SMB shares.

```
[OK] Completed: NSE smb-os-discovery on discovered service hosts  
[INFO] Starting: NSE smb-enum-shares on discovered service hosts  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-29 13:33 -0400  
Nmap scan report for 192.168.72.154  
Host is up (0.00067s latency).
```

ldap-rootdse - Retrieves Active Directory RootDSE information.

```
[OK] Completed: NSE smb-enum-shares on discovered service hosts  
[INFO] Starting: NSE ldap-rootdse on discovered service hosts  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-29 13:33 -0400  
Nmap scan report for 192.168.72.154  
Host is up (0.00079s latency).
```

Advanced Active Directory Enumeration

After completing the NSE scripts, the program performs advanced Active Directory enumeration using enum4linux and CrackMapExec (CME). These tools collect additional information about the domain, users, groups, security policies, and accessible SMB shares, providing a broader view of the target environment.

```
[OK] Completed: NSE ldap-rootdse on discovered service hosts
[INFO] Running Advanced AD Enumeration against target: 192.168.72.154
[INFO] Starting: Advanced Enumeration (enum4linux)
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Wed Jul 29 13:33:39 2026
```

```
[OK] Completed: Advanced Enumeration (enum4linux)
[INFO] Starting: SMB Share Enumeration (CME)
SMB      192.168.72.154 445      WIN-A8PBA9GL4GE [*] Windows Server 2016 Standard Evaluation 14393 x64 (
name:WIN-A8PBA9GL4GE) (domain:netsec.local) (signing:True) (SMBv1:True)
SMB      192.168.72.154 445      WIN-A8PBA9GL4GE [+] netsec.local\administrator:Password123! (Pwn3d!)
SMB      192.168.72.154 445      WIN-A8PBA9GL4GE [+] Enumerated shares
SMB      192.168.72.154 445      WIN-A8PBA9GL4GE
SMB      192.168.72.154 445      WIN-A8PBA9GL4GE
SMB      192.168.72.154 445      WIN-A8PBA9GL4GE
SMB      192.168.72.154 445      WIN-A8PBA9GL4GE ADMIN$      READ,WRITE      Remote Admin
SMB      192.168.72.154 445      WIN-A8PBA9GL4GE C$          READ,WRITE      Default share
SMB      192.168.72.154 445      WIN-A8PBA9GL4GE IPC$        Remote IPC
SMB      192.168.72.154 445      WIN-A8PBA9GL4GE NETLOGON    READ,WRITE      Logon server share
SMB      192.168.72.154 445      WIN-A8PBA9GL4GE ShareName   READ,WRITE
SMB      192.168.72.154 445      WIN-A8PBA9GL4GE SYSVOL      READ            Logon server share
[OK] Completed: SMB Share Enumeration (CME)
```

Basic Exploitation

The Basic Exploitation phase performs a controlled vulnerability assessment of the target environment. Before any exploitation-related actions are executed, the program verifies that the selected target is authorized and ensures that sensitive information, such as passwords, is not written to the output logs.

The module begins by executing an Nmap vulnerability scan. The collected results are stored in the session directory and included in the final report for later analysis.

[STEP] Exploitation

```
Selected exploitation level: Advanced  
Target/domain: 192.168.72.0/24 / netsec.local  
Password list path: wordlists/passwords.list  
Output is sanitized; passwords are not logged.
```

```
[INFO] Exploitation confirmed for authorized scope.
```

```
[INFO] Starting: Exploitation Basic vulnerability scan
```

```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-29 13:34 -0400
```

```
[OK] Completed: Exploitation Basic vulnerability scan
```

```
[INFO] Executing active Domain-wide Password Spraying attack...
```

```
[INFO] Starting: Password Spraying via CrackMapExec
```

Advanced Exploitation

The Advanced Exploitation phase extends the basic vulnerability assessment by performing additional Active Directory security validation tasks. These operations are executed in a controlled manner against the authorized target environment and provide deeper insight into domain security.

The workflow includes password spraying validation, Kerberoasting ticket requests, and offline Kerberos hash collection. The generated results are stored in the session directory for later analysis and reporting.

```
[OK] Completed: Password Spraying via CrackMapExec  
[INFO] Executing active Kerberoasting ticket request ...  
[INFO] Starting: Kerberoasting via Impacket (impacket-GetUserSPNs)  
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies
```

Offline Kerberos Hash Processing

```
[OK] Completed: Kerberoasting via Impacket (impacket-GetUserSPNs)  
[INFO] Executing active offline Kerberos hash cracking ...
```

Generated Reports

At the end of the assessment, the program automatically generates structured output files and a final execution summary. The reports include both raw scan results and parsed information, making it easier to review the collected data after the assessment has completed.

The generated reports are saved within the session directory in both text and PDF formats, providing a complete record of the executed assessment and its findings.

```
[STEP] Reporting
[INFO] Generating text report.
[OK] Text report saved: ./sessions/netsec.local_2026-07-29_13-32/reports/report.txt
[OK] PDF report saved with enscript and ps2pdf: ./sessions/netsec.local_2026-07-29_13-32/reports/report.pdf
[OK] Domain Mapper finished.
```

Summary

Target	192.168.72.0/24
Input domain	netsec.local
Domain provided	yes
Internal domain	netsec.local
Detected domain	netsec.local
Scanning	Basic
Enumeration	Advanced
Exploitation	Advanced
Hosts up	5
Possible DC	192.168.72.154 High confidence indicators: 53/tcp 88/tcp 389/tcp 445/tcp 3268/tcp
Vulnerabilities	high-risk findings found
Session	./sessions/netsec.local_2026-07-29_13-32
Text report	./sessions/netsec.local_2026-07-29_13-32/reports/report.txt
PDF report	./sessions/netsec.local_2026-07-29_13-32/reports/report.pdf

Generated PDF Report

In addition to the text report, the program automatically generates a PDF report that summarizes the assessment results in a structured and portable format. The report includes the execution summary, discovered hosts and services, enumeration findings, vulnerability assessment results, and references to the generated artifacts.

The PDF provides a convenient way to review, archive, or share the assessment results after the execution has completed.

Domain Mapper Executive Report

NETWORK SECURITY ASSESSMENT

Generated:	2026-07-29 15:35:15	Target Range:	192.168.72.0/24
Domain:	netsec.local (Detected: netsec.local)	Levels (S/E/X):	Basic / Advanced / Advanced

Report Findings & Summaries

Domain Mapper Report
=====

Metadata

Generated: 2026-07-29 15:35:15
Session Directory: ./sessions/netsec.local_2026-07-29_15-32
Safety Scope: Authorized lab use only

User Selections

Target Range: 192.168.72.0/24
Input Domain: netsec.local
User Provided Domain: yes
Internal Domain: netsec.local
Detected Domain: netsec.local
AD Username Provided: yes
AD Password Provided: yes
Password List Path: wordlists/passwords.list
Wizard Mode: yes
Scanning Level: Basic
Enumeration Level: Advanced
Exploitation Level: Advanced
Vulnerabilities: high-risk findings found

Assignment Coverage Checklist

Implemented:

- User input requirements, including optional hidden AD credentials and normalized levels.
- Basic scanning includes nmap -Pn and service detection (-sV).
- Intermediate scanning includes all TCP ports (-p-) with service detection (-sV).
- Advanced scanning includes UDP scanning of the top 100 UDP ports for practical runtime.
- Enumeration Basic reuses scanning service detection output instead of duplicating -sV.
- Enumeration Intermediate includes exactly three targeted NSE scripts.
- Exploitation Basic under the single YES-RUN-VULN confirmation.
- Wizard mode, help menu, session structure, logs, raw/parsed artifacts, text report, and PDF

Conclusion

The Domain Mapper project successfully automates the complete network assessment workflow, including user input collection, network scanning, service enumeration, controlled exploitation, and report generation. The modular design, structured output, and support for multiple operation levels provide a flexible and organized solution for network reconnaissance and analysis within authorized environments.

The project demonstrates how multiple open-source security tools can be integrated into a single automated workflow for efficient and organized network assessment.